

VR-Sets. Preprint.

Part VI. Cardinality, Cantor, Choice

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VI.0. Introduction

This part of the preprint is devoted to the most visible divergence of VR-Sets from classical ZFC: **the entire operational universe of VR-Sets is countable**. In classical ZF, already $\wp(\omega)$ is uncountable by Cantor's diagonal argument, and the universe V is a proper class of uncountable cardinality. In VR-Sets, $\wp(\omega)$ is countable (Part III §III.5), and this countability extends to the entire universe.

The divergence is not mathematical (the mathematics of countable and uncountable collections is identical in both systems under suitable formulation) but **ontological**: classical ZF admits the existence of objects without an operational description; VR-Sets admits only describable objects, and describable functionalities are countable in number.

In this part we:

- (1) precisely formulate and prove the theorem on the countability of the operational universe (§VI.1);
- (2) develop the analysis of Cantor's diagonal argument in the operational ontology — where it fails and why (§VI.2);
- (3) discuss the distinction between "countable from without" and "not enumerable from within" by analogy with computable analysis (§VI.3);
- (4) prove AC as a theorem on the countable universe (§VI.4);
- (5) fix the absence of the Banach–Tarski paradox in VR-Sets as a consequence of the absence of non-measurable sets (§VI.5);
- (6) discuss the methodological significance of a countable ontology — its relation to classical mathematics, constructivism, and computable analysis (§VI.6).

VI.1. Countability of the operational universe

Theorem VI.1

The collection of all operational sets of VR-Sets is countable.

Proof

Every operational set in VR-Sets is given by its description — a finite syntactic record of an operational procedure. Descriptions are written in a formal language: a fixed finite alphabet of symbols (the primitives $\emptyset$, $\rightarrow$, t from VR, together with auxiliary symbols — brackets,

commas, symbols for naming and referencing). A description is a finite string over this alphabet.

The set of all finite strings over a finite alphabet is countable: this is a basic fact of the theory of formal languages. Indeed, for each length n there are finitely many strings ($|\Sigma|^n$, where $|\Sigma|$ is the size of the alphabet), and a countable union of finite sets is countable.

Every operational set has a description (by the closure principle — Part II §II.3 — sets are precisely the describable functionalities). Consequently, the mapping "set $\rightarrow$ its description" is an injection from the collection of sets into the set of finite strings. The image of this injection is a subset of a countable set, hence itself countable.

The collection of operational sets of VR-Sets is countable.

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Remarks on the theorem

(1) The description is not unique. The same set may be given by different descriptions. For instance, $\emptyset$ may be described as "the empty operationality," as "the set revealing nothing," as O_0 , and so on. This means that the mapping "set $\rightarrow$ description" is many-to-one, but the injection in the other direction — "description $\rightarrow$ the set it specifies" — is well-defined (up to $\equiv$). The collection of sets is the image of a countable collection of descriptions modulo extensionality $\equiv$, and the image of a countable set is countable.

(2) Descriptions are countable even for infinite sets. ω is infinite in operational depth, but the description of ω is finite: "the n -th query yields O_n ." $\mathbb{R}_{\text{VR-Sets}}$ contains infinitely many reals, but each of them is given by a finite algorithm. The finiteness of the description is not in contradiction with the infinity of what it describes; the description is a syntactic record of a rule, and a rule can generate infinitely many consequences.

(3) Comparison with classical ZF. In classical ZF, the sentences of the language are also countably many (Löwenheim–Skolem), but the *objects* in the standard model are uncountable — because the classical theory admits the existence of objects not specified by any sentence. VR-Sets has no such gap between sentences and objects: each object is a describable functionality, and this relation is constitutive, not a property that objects may or may not have.

VI.2. Cantor's diagonal argument in the operational ontology

The most famous proof of uncountability is Cantor's diagonal argument for $\wp(\omega)$ and for $\mathbb{R}$. In classical ZF, it constructs from any proposed enumeration of subsets of ω a new subset distinct from all those enumerated, and concludes that no such enumeration is complete.

In VR-Sets $\wp(\omega)$ is countable (Part III §III.5). This means that the diagonal argument in the operational ontology **does not produce a new set**. We analyse precisely where it ceases to operate.

The classical argument

Suppose we have an enumeration $S_0, S_1, S_2, \dots$ of all subsets of ω . Define:

$$D = \{ n \in \omega : n \notin S_n \}.$$

The set D differs from each S_n at the element n : D contains n if and only if S_n does not contain n . Hence $D \neq S_n$ for every n . D is a new subset of ω not appearing in the enumeration, contradicting the completeness of the enumeration. Cantor's conclusion: no complete enumeration exists, $\wp(\omega)$ is uncountable.

What the classical argument presupposes

The argument uses two premises:

- (a) **Completeness of the enumeration:** $S_0, S_1, \dots$ covers all subsets of ω .
- (b) **D is a well-defined set:** the condition " $n \notin S_n$ " specifies a subset of ω .

In classical ZF both premises are accepted without reservation. In the operational ontology each of them requires verification.

Where the argument fails in VR-Sets

The obstacle is in premise (a), and it is deep.

A complete enumeration of describable subsets of ω is operationally unrealisable. Every subset of ω in VR-Sets is an operationally describable functionality. The collection of such functionalities is countable (Theorem VI.1). However, *enumerating* them in an operationally describable sequence $S_0, S_1, \dots$ is a different task. A description of such an enumeration requires that for every n there exist an operational procedure returning the n -th subset on the list. This procedure must:

- (i) enumerate all finite descriptions over the alphabet of operational procedures — this is describable;
- (ii) **verify that each description specifies a correct functionality** — that is, terminates on every query, gives a definite response, is not self-contradictory. This is an **undecidable task**: the question of whether an arbitrary description is correct is equivalent to the halting problem.

Consequently, an *operationally describable* enumeration of all describable subsets of ω does not exist. Premise (a) of the diagonal argument is unrealisable in the operational ontology.

What this implies for D

The definition $D = \{ n \in \omega : n \notin S_n \}$ requires access to S_n for each n — that is, it requires an operational enumeration. If no such enumeration exists, the description of D does not specify an operational functionality: on the query " $n \in D$?" the procedure would have to refer to S_n , which is undefined by the operational enumeration.

By the closure principle (Part II §II.3), a description that does not specify an operationally describable functionality does not give rise to a set. **D is not a set in VR-Sets.** The diagonal argument in the operational ontology is not a proof of uncountability but a demonstration that the attempt to construct a complete enumeration encounters the halting problem.

Relation to the classical result

Cantor's classical result remains true: in classical ZF, $\wp(\omega)$ is uncountable. VR-Sets does not refute Cantor — it works on a different ontological basis. The difference may be formulated thus:

- **In classical ZF**, the existence of subsets of ω not specified by any description is admitted. Such subsets "are" in the sense of classical ontology and form an additional uncountability beyond the describable.
- **In VR-Sets**, there are no indescribable subsets: a set is a description, full stop. Cantor's uncountability is an artefact of admitting indescribable objects; in the operational ontology that admission is not made, and uncountability does not arise.

This is consistent with the known observation of Skolem (the Skolem paradox): every set theory in a countable language has a countable model, and "uncountability" within such a model is a relative notion. VR-Sets is not simply a countable model of ZFC (Part IV §IV.5), but an ontologically coherent alternative in which the countability of the operational universe is the natural state, not paradoxical.

VI.3. Countable from without vs. not enumerable from within

A subtle distinction important for the understanding of the status of countability in VR-Sets. It is well known in computability theory and computable analysis; VR-Sets shares this distinction and relies on it.

Setting

A collection X is called **countable from without** if it is countable as an object of classical set theory — that is, there exists a bijection between X and $\mathbb{N}$ in classical ZF.

A collection X is called **enumerable from within** (recursively enumerable) if there exists an operational procedure that, as its n -th output, returns the n -th element of X .

These two notions **do not coincide**. There exist countable collections that are not recursively enumerable. A classical example from computability theory is the set of indices of terminating programs: it is countable (a subset of $\mathbb{N}$) but not recursively enumerable (by Rice's theorem).

Application to VR-Sets

The collection of all describable subsets of ω in VR-Sets:

- **Countable from without** (Theorem VI.1): descriptions are countable, sets are countable.
- **Not enumerable from within**: there is no operational procedure that, for every n , returns the n -th describable subset of ω .

This distinction dissolves the apparent tension: "if $\wp(\omega)$ is countable, why cannot one exhibit an enumeration?" The answer: countability does not imply enumerability. The set exists as a countable object; but a procedure that would list its elements does not exist.

Analogy with computable reals

The same situation is well known in computable analysis for the field of computable reals:

- The field of computable reals is countable (a subfield of the classical $\mathbb{R}$, consisting of numbers with an operational description).
- However it is not enumerable: the halting problem prevents the separation of terminating algorithms for computing reals from non-terminating ones.

$\mathbb{R}_{\text{VR-Sets}}$ stands in the same position: a countable subfield of the classical $\mathbb{R}$, isomorphic to the field of computable reals, not enumerable from within. This is consistent with the position of VR-Numbers and confirms that VR-Sets and computable analysis work on the same ontology — the one in which "operational" means "algorithmically specifiable."

Where this is especially important

The distinction "from without vs. from within" dissolves several apparent problems:

- (a) *Cantor's diagonal* requires enumerability from within; countability from without is insufficient. Since $\wp(\omega)$ in VR-Sets is countable from without but not enumerable from within, the diagonal does not construct D.
- (b) *Full AC on the entire universe* does not reduce to simple "choose one element from each set of a countable family" — an operational procedure of choice is required, and for an arbitrary family it may not be describable. See §VI.4 for details.
- (c) *The Löwenheim–Skolem theorem* predicts the existence of countable models of ZFC from without; in VR-Sets such a model is countable also from within (though not its entire structure is enumerable from within). This is consistent with Conjecture IV.1.

VI.4. The axiom of choice as a theorem

Part III §III.9 has already established: countable AC is a theorem of VR-Sets, and full AC on the countable universe is equivalent to the countable one. Here we develop this result, examine the connection with DC (dependent choice), and fix what distinguishes the VR-Sets position from the standard treatment of choice in ZFC.

Three levels of choice

In classical set theory three forms of the axiom of choice are distinguished:

- **Countable Choice (CC):** for a countable family of non-empty sets there exists a choice function. Equivalent to countable AC, without which many basic facts of analysis cannot be proved (for instance, that a countable union of countable sets is countable).
- **Dependent Choice (DC):** for every binary relation R on a non-empty set A such that for every $x \in A$ there exists y with xRy , there exists a sequence $x_0, x_1, x_2, \dots$ with $x_n R x_{n+1}$ for all n . Stronger than CC.
- **Full AC:** for an arbitrary family of non-empty sets there exists a choice function. Stronger than DC; the source of the Banach–Tarski paradox.

Status of each level in VR-Sets

DC is built into the operational ontology. Every operational procedure consists of specific steps: at each step the procedure makes a definite choice, and subsequent steps depend on those already made. This is precisely the structure of DC. In the operational ontology, DC is not an axiom but a property of operational action itself: it is impossible to conceive of an operational procedure that does not make dependent choices.

Technically: for a binary relation R given by a describable procedure, a sequence $x_0, x_1, x_2, \dots$ is constructed operationally — choosing at each step the first element satisfying R with the preceding one. "First" here refers to the order of enumeration given by the description of R . This sequence is itself operationally describable and, by the closure principle, is a set.

Countable AC is a theorem (§III.9). The proof in §III.9 showed: for a countable family of non-empty sets, the choice function is operationally describable (on the i -th query, reveal the first element of A_i). By the closure principle, this is a set.

Full AC is a theorem on the countable universe. Since the entire universe of VR-Sets is countable (Theorem VI.1), every "family of non-empty sets" is a countable family. Hence full AC reduces to countable AC, which is a theorem.

Theorem VI.2 (AC in VR-Sets)

In VR-Sets the axiom of choice in its full form is a theorem: for every family of non-empty sets there exists an operational choice function.

Proof. The collection of sets in VR-Sets is countable (Theorem VI.1). A family of non-empty sets is an operational set whose functionality reveals the elements of the family. By countability the whole family has a description of the form $A_0, A_1, A_2, \dots$ (a countable enumeration). The choice function: $\text{choice}(i) = \text{the first element of } A_i$ — is a describable procedure. By the closure principle, choice is a set.

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Subtlety: the operability of choice

The formulation "the choice function choice" in this proof depends on the *order of presentation* of the elements of A_i and A_{i+1} : "the first element of A_i " is the element revealed by the functionality A_i on its first query. Different presentations may give different "first elements," and then choice is a different functionality (formally — several different sets choice).

There is nothing paradoxical here: the operational theorem says that *some* choice function exists, not that it is unique. This is consistent with the classical situation: AC in ZFC asserts the existence of a choice function, not its uniqueness.

Where VR-Sets differs from ZFC on AC

Substantive difference. In classical ZFC, full AC is an independent axiom. From ZF without AC one can build models in which AC is false, and these models are mathematically substantive (for instance, models with the axiom of determinacy AD). In VR-Sets there is no such gap: AC holds automatically by virtue of the countability of the universe. To reject AC

in VR-Sets would mean to reject countability, and countability is a consequence of operationality, not a separate postulate.

Methodological difference. In ZFC, AC is often regarded as "suspect" because of its non-constructive consequences such as Banach–Tarski. In VR-Sets, AC is harmless, because the objects on which ZFC builds paradoxical consequences (non-measurable subsets of $\mathbb{R}$, ultrafilters on $\mathbb{N}$) do not exist in the operational ontology. See §VI.5.

VI.5. Absence of the Banach–Tarski paradox

The most famous "pathological" consequence of AC is the Banach–Tarski theorem: the unit ball in $\mathbb{R}^3$ can be partitioned into a finite number of pieces, and from these pieces two unit balls can be assembled. The proof makes essential use of full AC to construct non-measurable subsets of the sphere.

In VR-Sets the Banach–Tarski paradox **is absent**. We fix precisely why — because this is the most graphic point where the operational ontology departs from the classical one not only formally but substantively.

Where Banach–Tarski requires AC

The Banach–Tarski proof consists of several steps; the key one uses AC to construct a **non-measurable Vitali subset** of the sphere. A Vitali set is a set of representatives of equivalence classes on the sphere under an appropriate relation. The existence of such a set requires the selection of one representative from an uncountable family of classes — which is an application of full AC for uncountable cardinality.

Once the non-measurable set has been constructed, the rest of the proof combines the free group of rotations of $\mathbb{R}^3$ with this set, obtaining a partition of the ball into a finite number of pieces and a reconstruction of two balls.

Why this does not happen in VR-Sets

Each of the steps relies on classical objects absent in VR-Sets:

- (1) **Uncountable family of equivalence classes on the sphere.** In classical ZFC, the sphere S^2 is uncountable (as a subset of $\mathbb{R}^3$, classically uncountable), and a factor under a suitable relation gives an uncountable family. In VR-Sets there is no classically-uncountable sphere: $S^2_{\text{VR-Sets}}$ is a countable set of describable points, and any factor remains countable. No uncountable family of equivalence classes arises.
- (2) **Non-measurable Vitali set.** The Vitali construction requires the selection of one representative from an uncountable family of classes. In VR-Sets there are no such families; countable AC gives a choice function without any non-measurability.
- (3) **Classical $\mathbb{R}^3$.** The geometric arguments of Banach–Tarski operate on classical $\mathbb{R}^3$ — an uncountable space with the full apparatus of Lebesgue measure and non-measurable sets. $\mathbb{R}_{\text{VR-Sets}}$ is isomorphic to the field of computable reals and countable; $\mathbb{R}^3_{\text{VR-Sets}}$ is also countable. On a countable set, measure behaves standardly (countably additive), and non-measurable subsets do not arise.

Assertion VI.1

The Banach–Tarski paradox is not reproduced in VR-Sets. This follows not from a weakening of AC (AC in VR-Sets is a theorem) but from the absence of classically-uncountable objects on which the paradox is built.

Methodological commentary

The Banach–Tarski paradox is often regarded as "evidence of the pathology" of full AC in classical mathematics. The standard response is that this is the price of working with the uncountable continuum and its subsets without operational restrictions.

VR-Sets gives a different response: the paradox arises not from AC, but from the attempt to combine AC with the admission of classically-uncountable objects lacking an operational description. If the ontology is restricted to operational objects, AC is a harmless theorem, and there are no paradoxes. The difference is not in the mathematical apparatus, but in the ontological commitments.

This is consistent with the position of Solovay, Shelah, and others in work on models of ZF + DC: in such models AD is compatible with DC, there are no non-measurable sets, and analysis works without paradoxes. VR-Sets is the operational reformulation of this mathematical line: what is achieved in Solovay's models by restricting the universe (only definable sets) is, in VR-Sets, the natural state of the ontology.

VI.6. Methodological significance of a countable ontology

Closing the part, we discuss how the countability of the operational universe relates to classical mathematics and other alternative approaches. This is a methodological section and contains no new theorems.

What VR-Sets loses

Direct losses in comparison with classical ZFC:

- **Uncountable cardinalities:** $\aleph_1$, $\aleph_2$, ... and the whole theory of uncountable ordinals are absent in VR-Sets as separate ontological levels. Cardinality in VR-Sets takes only two values: "finite" and "countable."
- **The continuum hypothesis:** the question "is the cardinality of $\mathbb{R}$ greater than the cardinality of $\mathbb{N}$?" does not arise in VR-Sets, because both cardinalities are countable.
- **Uncountable classical objects:** the set of all functions $\mathbb{N} \rightarrow \mathbb{N}$, the set of all subsets of $\mathbb{R}$, equivalence classes with an uncountable number of representatives — all such objects *in their classical form* are absent. Their operational analogues — describable functions, describable subsets — exist and are countable.

What VR-Sets retains

Direct retentions:

- **All of arithmetic and number theory** (Part V).

- **All of computable mathematics:** algorithms, recursive functions, formal languages, computability theory — all of this works with operational objects by construction.
- **Analysis on computable reals:** $\mathbb{R}_{\text{VR-Sets}}$ supports integration, differentiation, power series — everything that does not require the existence of non-measurable sets or uncountable cardinalities.
- **Constructive and reverse mathematics:** systems such as RCA_0 , WKL_0 , ACA_0 — all formalisable in the countable ontology of VR-Sets.
- **Algebra and combinatorics on countable structures:** the theory of groups, rings, fields, graphs — all of this for countable carriers works without modification.

Relation to other approaches

VR-Sets shares its ontological position with several well-known approaches in the foundations of mathematics:

- **Computable analysis** (Weihrauch, Pour-El–Richards): mathematics on computable objects, including computable reals. The operational universe of VR-Sets is isomorphic to the basic structure of computable analysis.
- **Constructivism** (Brouwer, Bishop): mathematics on constructively given objects. VR-Sets is close to this position ontologically, but not necessarily methodologically (for instance, classical logic in VR-Sets operates without restrictions on the law of excluded middle).
- **Predicativism** (Poincaré, Weyl, Feferman): mathematics on predicatively definable objects. VR-Sets is ontologically still more restrictive (only operationally describable objects), but substantively close.
- **Solovay, ZF + DC + AD:** models of ZF with dependent choice and the axiom of determinacy, without full AC and without non-measurable sets. VR-Sets gives an operational ontological formulation of this mathematical position.

The differences of VR-Sets from each of these approaches lie in the manner of formulation. All the listed approaches construct countable or near-countable mathematics by *restricting* the classical universe. VR-Sets constructs it by *ontological choice*: countability is a consequence of the operational ontology, not a restriction imposed on the classical one.

Concluding commentary

The countability of the operational universe of VR-Sets is not a defect relative to classical mathematics but an ontological position consistent with the slogan "only $\emptyset$ is, all else is doing." All objects of VR-Sets are actions, and operationally describable actions are countable in number. Classical mathematics not fitting this picture (uncountable cardinalities, non-measurable sets, paradoxes of choice) works on a different ontology — the one in which the existence of objects without an operational description is admitted.

Which of the two ontologies is "correct" is not a mathematical question. Both ontologies are internally non-contradictory; both are fruitful. VR-Sets is the explicit fixing of the operational ontology, with all its consequences. Part VII will discuss how this position interacts with self-referential statements and Tarski's theorem on the undefinability of truth.

VI.7. Summary

In this part we have established:

- (1) **Theorem VI.1**: the collection of all operational sets of VR-Sets is countable. Proof — through an injection into the set of finite descriptions over a finite alphabet.
- (2) **Cantor's diagonal argument does not operate in VR-Sets** (§VI.2): its premise of a complete enumeration of describable subsets is operationally unrealisable (equivalent to the halting problem), and so the diagonal D is not a set.
- (3) **Countable from without $\neq$ enumerable from within** (§VI.3): $\wp(\omega)$ in VR-Sets is countable as an object but not enumerable by procedure. Analogy with computable reals.
- (4) **Theorem VI.2** (§VI.4): full AC in VR-Sets is a theorem, a consequence of the countability of the universe. DC is built into operability; countable AC was already proved in Part III §III.9.
- (5) **Assertion VI.1** (§VI.5): the Banach–Tarski paradox is absent in VR-Sets, not because of a weakening of AC but because of the absence of classically-uncountable objects on which the paradox is built.
- (6) **Relation to other approaches** (§VI.6): VR-Sets is ontologically close to computable analysis, constructivism, predicativism, and models of ZF + DC + AD; it differs in the manner of formulation — by ontological choice, not by restriction of the classical universe.

What follows. Part VII treats Tarski's theorem on the undefinability of truth in the light of the operational ontology and connects it with the duality of axiom A1 from VR. This is the most philosophically loaded part of the preprint; the formulation is cautious — it reformulates, it does not resolve.